

Spectral Theory and Hilbert-Space Embedding of the FFGFT Tori

Mathematical Foundation of the $T^4/\mathbb{Z}_3$ Spectral Mechanics

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Summary: the particle masses are eigenvalues of the mass operator $\hat{M}_f$, not of $\hat{F}$; ξ remains the smallest eigenvalue of the D_4 sub-operator of $\hat{F}$.
- Recursive measure: note – the scheme yields $n^{50.5\xi}$ and $D_f^{(100)} = 3$, not n^ξ .
- $\mathbb{Z}_3$ projector: $\frac{1}{3} \sum \omega^k \tau^k \rightarrow \frac{1}{3} \sum \tau^k$ (invariant subspace).
- Electron eigenvalue (lepton table, correspondence table): $0.511 \rightarrow 0.505$ MeV (formula value).
- Neutrino table, ν_1 : $\dim_{\text{loc}} = 2 - 1/4 \rightarrow 2 + 1/4$.
- Selection rule: [K] “spectral theorem” $\rightarrow$ [S] orbit structure; $\Delta p = k/3$ holds only within one orbit.
- Self-adjointness (1 Oct. 2026): restricted to the $\mathbb{Z}_3$ -invariant sector P_0 (Doc. 327, R145); $m_\tau 1783.5 \rightarrow 1783.4$ MeV.

List of Symbols

Symbol	Meaning	Value / Unit
$\mathcal{H}$	Hilbert space of physical states	separable
$\hat{F}$	fundamental fractal spectral operator	self-adjoint
$\hat{F}_{D_4}$	D_4 sub-operator	eigenvalue ξ
$\sigma(\hat{F})$	spectrum of $\hat{F}$	$\subset \mathbb{R}$
$E(\lambda)$	spectral projection measure	$E(\lambda)^2 = E(\lambda)$
$\mathcal{D}(\hat{F})$	domain of $\hat{F}$	dense in $\mathcal{H}$
$P_{\mathbb{Z}_3}$	$\mathbb{Z}_3$ projection operator	idempotent
$\hat{\Delta}_{T^4}$	Laplacian on $T^4/\mathbb{Z}_3$	essentially self-adj.
$\hat{W}$	winding-number operator	eigenvalues (n_θ, n_ϕ)
D_f	fractal dimension of the orbifold	$3 - \xi$
D_f^{frac}	approximate fractal dimension	2.94 (no ξ -derivation)
K_{frac}	interface factor rolled $\rightarrow$ SI	$1 - 100\xi = 74/75$ (Doc. 133)
ξ	FFGFT geometry parameter	$4/30000$
ω	$\mathbb{Z}_3$ phase factor	$e^{2\pi i/3}$
[K]	derivable from ξ	status marker
[B]	algebraically proved	status marker
[S]	sketched / plausible	status marker
[SETZUNG]	axiomatic setting	status marker

Summary

Doc. 322 develops the mathematical foundation of FFGFT spectral theory. The mass formulae of Doc. 320 receive here their rigorous Hilbert-space embedding: the mass operator $\hat{M}_f$ on the state space $\mathcal{H}_{\text{FFGFT}}$ has the particle masses as eigenvalues; the geometry parameter $\xi = 4/30000$ is the smallest eigenvalue of the D_4 sub-operator of $\hat{F}$. (*Corrected on 1 Oct. 2026: previously “the fundamental fractal operator $\hat{F}$... has the particle masses*

as eigenvalues” – the derivation in Section .7 runs through the mass operator $\hat{M}_f$; by the definition of its domain $\hat{F}$ is bounded through $\sum_n r_n$ (Doc. 327, R86) and does not carry the mass hierarchy.) The mass formulae of Docs. 006, 317 and 320 are not phenomenological ansätze but eigenvalue equations of a self-adjoint operator.

Status markers

All statements carry FFGT status markers:

[K] derivable from ξ **[B]** algebraically proved **[S]** sketched/plausible **[SETZUNG]** axiomatic setting

.1 Foundations of Spectral Theory in Hilbert Spaces

Motivation: from matrices to operators

In linear algebra a symmetric $n \times n$ matrix A possesses exactly n real eigenvalues and an orthonormal eigenbasis, enabling the diagonal decomposition $A = UDU^T$.

In quantum mechanics – and in FFGT – operators act on *infinite-dimensional* Hilbert spaces. The momentum operator, the Hamiltonian, and the FFGT spectral operator have no finite-dimensional analogue. Spectral theory generalises the eigenvalue concept to this case.

The Hilbert space: definition and physical meaning

Definition (Hilbert space). A Hilbert space $\mathcal{H}$ is a complete inner-product vector space. Its elements have finite norm $\|\psi\| = \sqrt{\langle\psi, \psi\rangle} < \infty$, which in position space means

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty. \quad (1)$$

Only such square-integrable wave functions are physically realisable – they describe normalisable probability densities.

Note: Eigenstates of operators with *continuous* spectrum – e.g. position eigenstates $|y\rangle$ with $\langle x|y\rangle = \delta(x - y)$ – are not normalisable and lie outside $\mathcal{H}$ itself, in the extended distributional space. In FFGT this issue arises in the IR limit; the sub-Planck regime has a purely discrete spectrum.

The spectrum of a linear operator

For a linear operator T on a Hilbert space the spectrum $\sigma(T)$ generalises the set of all eigenvalues:

$$\sigma(T) = \{\lambda \in \mathbb{C} : (T - \lambda I) \text{ not invertible with bounded inverse}\}. \quad (2)$$

One distinguishes:

- **Point spectrum** $\sigma_p(T)$: “true” eigenvalues with $T\psi = \lambda\psi$, $\psi \neq 0$.
- **Continuous spectrum** $\sigma_c(T)$: $(T - \lambda I)$ injective but not surjective; no normalised eigenvector.
- **Absolutely continuous and singular spectrum**: important for Schrödinger operators on fractal spaces.

The spectral theorem: centrepiece of the theory

[B] For self-adjoint operators ($T = T^\dagger$ with suitable domain) the spectral theorem holds: there exists a unique *spectral measure* E – a projection-valued function $\lambda \mapsto E(\lambda)$ – such that

$$T = \int_{\sigma(T)} \lambda dE(\lambda). \quad (3)$$

For compact operators Eq. (3) reduces to the familiar discrete sum $T = \sum_i \lambda_i \langle \cdot, e_i \rangle e_i$. Self-adjointness guarantees $\sigma(T) \subset \mathbb{R}$ – real spectra, the necessary condition for measurable observable values.

.2 The Fundamental Fractal Operator $\hat{F}$

Definition and Hilbert-space structure

[SETZUNG] FFGFT postulates a quantum-logical fractal operator $\hat{F}$ acting on a separable Hilbert space $\mathcal{H}$, formally encoding the fractal self-similarity of the sub-Planck geometry:

$$\hat{F} \equiv \bigoplus_{n=1}^{\infty} S_n \circ L_n, \quad (4)$$

with scale operators $S_n = r_n U_n$ ($r_n \in (0, 1)$ contraction rates, U_n unitary) and logical projectors L_n satisfying $L_n^2 = L_n$, $L_n^\dagger = L_n$, $L_n L_m = L_{\min(n,m)}$.

The domain

$$\mathcal{D}(\hat{F}) = \left\{ \psi \in \mathcal{H} : \sum_{n=1}^{\infty} \|S_n(L_n \psi)\|^2 < \infty \right\} \quad (5)$$

is dense in $\mathcal{H}$. **[B]** $\hat{F}$ is self-adjoint: since $L_0 = \xi \cdot \ell_P$ (Doc. 180) renders the scale ladder finite, $\sum_n r_n < \infty$ and hence $\hat{F}$ is bounded with $\mathcal{D}(\hat{F}) = \mathcal{H}$; symmetry follows from the $\mathbb{Z}_3$ pairing $(k, -k)$ on the $\mathbb{Z}_3$ -invariant sector P_0 . The deficiency indices are $(0, 0)$ – there is exactly one self-adjoint realisation (Doc. 327, R86). *(Corrected on 1 Oct. 2026: previously without the restriction to P_0 – by Doc. 327, Theorem 3 (corrected), $\hat{F}$ is normal under the pairing, $\varphi_{-k} = \bar{\varphi}_k$; it is self-adjoint on P_0 , on the full space only $\text{Re } \hat{F} = (\hat{F} + \hat{F}^\dagger)/2$ as a choice; R145.)*

ξ as smallest eigenvalue of the D_4 sub-operator

[K] The geometry parameter is not a free parameter but a spectral value:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}. \quad (6)$$

This explains structurally why ξ appears as the base of all mass formulae: it is not a chosen parameter but the lowest mode of the geometry operator.

.3 The FFGFT State Space

Tensor-product structure

[K] The physical state space:

$$\mathcal{H}_{\text{FFGFT}} = \mathcal{H}_{\text{geom}} \otimes \mathcal{H}_{\text{spin}} \otimes \mathcal{H}_{\text{flavor}}, \quad (7)$$

with:

1. $\mathcal{H}_{\text{geom}} = L^2(T^4/\mathbb{Z}_3, d\mu_f)$: square-integrable functions on the fractal orbifold.
2. $\mathcal{H}_{\text{spin}} = \mathbb{C}^{2[D_f]}$: spinor space; $[D_f] = 3$.
3. $\mathcal{H}_{\text{flavor}} = \ell^2(\mathbb{N}^2)$: flavour space with winding numbers (n_θ, n_ϕ) .

The fractal measure $d\mu_f$ with 100-fold recursion

[K] The fractal measure on $T^4/\mathbb{Z}_3$ is built by 100 successive fractalisations of the Lebesgue measure:

$$d\mu_f^{(0)}(x) = dx, \quad (8)$$

$$d\mu_f^{(k+1)}(x) = \prod_{i=1}^4 \left(\sum_{n=1}^{100} c_n^{(k)} \delta(x_i - x_i^{(n)}) \right) d\mu_f^{(k)}(x), \quad (9)$$

with $c_n^{(k)} = n^{-(D_f^{(k)}-3)}$, $D_f^{(k)} = 3 - \xi^{(k)}$, $\xi^{(k)} = (4/30000)(1 - k/100)$. After 100 steps:

$$\boxed{d\mu_f(x) = \prod_{i=1}^4 \left(\sum_{n=1}^{100} n^\xi \delta(x_i - x_i^{(n)}) \right) dx_i}, \quad \xi = \frac{4}{30000}. \quad (10)$$

(Note of 1 Oct. 2026: according to the scheme for $\xi^{(k)}$, $\xi^{(100)} = 0$ and hence $D_f^{(100)} = 3$. The product of the coefficients $c_n^{(k)} = n^{\xi^{(k)}}$ over $k = 0, \dots, 99$ gives $n^{\xi \sum_k (1-k/100)} = n^{50.5\xi} \approx 1 + 0.0067 \ln n$ (about 1.031 at $n = 100$), not n^ξ ; the boxed measure corresponds only to the first step $k = 0$. $D_f = 3 - \xi$ as a limit does not follow from the recursion in this form.)

.4 The Torus T^4 in Hilbert Space

Fourier decomposition on T^4

[B]

$$L^2(T^4) \cong \bigoplus_{n \in \mathbb{Z}^4} \mathbb{C} \cdot e^{in \cdot x/R}, \quad \Delta_{T^4} e^{in \cdot x/R} = -\frac{|n|^2}{R^2} e^{in \cdot x/R}. \quad (11)$$

The $\mathbb{Z}_3$ orbifold projection

[K]

$$P_{\mathbb{Z}_3} = \frac{1}{3} \sum_{k=0}^2 \tau^k, \quad \mathcal{H}(T^4/\mathbb{Z}_3) = \text{Fix}(P_{\mathbb{Z}_3}) \subset L^2(T^4). \quad (12)$$

A state lies in the orbifold space exactly when $\tau\psi = \psi$. (Corrected on 1 Oct. 2026: previously $P_{\mathbb{Z}_3} = \frac{1}{3} \sum_k \omega^k \tau^k$ – for $\tau\psi = \lambda\psi$ one has $\frac{1}{3} \sum_k (\omega\lambda)^k \psi = \psi$ only for $\lambda = \omega^{-1}$, otherwise 0; the old operator projects onto the charged sector $\tau\psi = \omega^{-1}\psi$, the invariant projector is $\frac{1}{3} \sum_k \tau^k$. The fixed-point condition for the neutrinos below is to be read with this projector.)

Fixed points and neutrino localisation

[K] The boundary condition at a fixed point x_0 :

$$\psi_\chi(x_0 + y) = \chi \cdot \psi_\chi(x_0 + g_*(y)). \quad (13)$$

The three neutrino flavours are localised at fixed points (Doc. 320):

$$\nu_1 \leftrightarrow F_2 = \frac{2\pi R}{3}(1, 1, 1, 0), \quad \nu_2 \leftrightarrow F_5 = \frac{4\pi R}{3}(1, 1, 1, 0), \quad \nu_3 \leftrightarrow F_7 = \frac{4\pi R}{3}(1, 1, 1, 1). \quad (14)$$

.5 The Complete Spectral Operator

Tensor-product form

[K]

$$\hat{F} = \hat{\Lambda}_{T^4} \otimes \hat{S} \otimes \hat{W}, \quad \hat{M}_f = v \cdot \xi \hat{P} \otimes \hat{r}. \quad (15)$$

Commutator conditions and MASA

[K]

$$[\hat{F}, \hat{S}] = 0, \quad [\hat{n}_\theta, \hat{n}_\phi] = 0. \quad (16)$$

The MASA $\mathcal{A} = \text{span}\{\hat{n}_\theta, \hat{n}_\phi, \hat{p}\}$ yields the complete spectral labelling (n_θ, n_ϕ, p) .

.6 Generalised Fourier Transform on $T^4/\mathbb{Z}_3$

[K] Projection kernel and GFT:

$$K_\lambda^{(p)}(x) = \sum_{n \in \mathbb{Z}^4} \frac{\delta_{\lambda, \Lambda_n^{(p)}}}{|n|^{D_f/2}} e^{in \cdot x/R}, \quad \Lambda_n^{(p)} = \frac{|n|^2}{R^2} \cdot \xi p. \quad (17)$$

$$\tilde{\psi}(\lambda) = \langle K_\lambda^{(p)}, \psi \rangle_{\mathcal{H}} = \int_{T^4/\mathbb{Z}_3} \overline{K_\lambda^{(p)}(x)} \psi(x) d\mu_f(x). \quad (18)$$

.7 Spectral Derivation of Particle Masses

Charged leptons as torus excitations

[K] The mass eigenvalue equation $\hat{M}_{e,\mu,\tau}|n_\theta, n_\phi, p\rangle = m_i|n_\theta, n_\phi, p\rangle$ gives:

$$m_i = \frac{n_\phi^2}{n_\theta} \cdot \xi^{p_i} \cdot v. \quad (19)$$

Particle	n_θ	n_ϕ	p_i	m_i
Electron e	3	2	3/2	0.505 MeV
Muon μ	5	4	1	104.96 MeV
Tau τ	9	5	2/3	1783.4 MeV

(Corrected on 1 Oct. 2026: previously electron 0.511 MeV – the row showed the measured value instead of the formula value; $(4/3)\xi^{3/2} \cdot 246 \text{ GeV} = 0.5050 \text{ MeV}$ (with $v = 246.22 \text{ GeV}$: 0.5054 MeV), –1.2 % below the measured 0.511 MeV. The muon row 104.96 MeV is the formula value with $v = 246 \text{ GeV}$.)

Neutrinos as fixed-point eigenstates

[K]

$$P_{\mathbb{Z}_3}|v_i\rangle = |v_i\rangle, \quad m_{v_i} = m_e \cdot \xi^{p_i}. \quad (20)$$

Neutrino	Fixed point	$\dim_{\text{loc}}$	p_i	m_{ν_i}
ν_1	F_2	$2 + 1/4$	$9/4$	0.976 MeV
ν_2	F_5	2	2	9.084 MeV
ν_3	F_7	$2 - 1/5$	$9/5$	54.11 MeV

(Corrected on 1 Oct. 2026: previously $\dim_{\text{loc}} = 2 - 1/4$ for $\nu_1 - 2 - 1/4 = 7/4 \neq p_1 = 9/4$; the mass $m_e \xi^{9/4} = 0.976 \text{ MeV}$ belongs to $p = 9/4$ (with $7/4$ it would be 84.5 MeV); for ν_2 and ν_3 , $p = \dim_{\text{loc}}$ holds.)

Spectral selection rule for the exponent ladder

[S] The commutator condition

$$\hat{F} \cdot \hat{\xi}^{1/3} = \hat{\xi}^{1/3} \cdot \hat{F} \quad (21)$$

makes the spectrum invariant under scaling by $\xi^{1/3}$. Exponents on the same orbit differ by multiples of $1/3$:

$$\Delta p = \frac{k}{3}, \quad k \in \mathbb{Z} \quad (\text{within one orbit}), \quad (22)$$

as for $\mu \rightarrow \tau$ ($\Delta p = 1/3 = 1/D$, Doc. 158), with ν_2 on the same orbit. The exponent ladder spans several orbits: p_e , p_{ν_1} and p_{ν_3} do not lie on the $1/3$ grid of the muon orbit. The rule therefore describes an *orbit structure* of the operator $\hat{F}$; the commutator condition itself is not derived. (Corrected on 1 Oct. 2026: previously [K] $\Delta p = k/3$ as a spectral theorem for all exponents – $3/2 - 1 = 1/2$, $9/4 - 2 = 1/4$, $2 - 9/5 = 1/5$ are not multiples of $1/3$, only $1 - 2/3 = 1/3$ fits; the commutator condition does not enforce a ladder across all exponents.)

.8 Fractal Dimension and K_{frac}

[K] Exact FFGFT formula:

$$D_f = 3 - \xi = 3 - \frac{4}{30000} = \frac{22499}{7500} \approx 2.9998\bar{6}. \quad (23)$$

Doc. 133 gives the interface factor:

$$K_{\text{frac}} = 1 - 100\xi = \frac{74}{75} \approx 0.9866\bar{6}. \quad (24)$$

In mass ratios K_{frac} cancels exactly (Doc. 012).

.9 Complete Spectral Correspondence Table

(Corrected on 1 Oct. 2026: previously $|e\rangle$ eigenvalue 0.511 MeV – formula value $\frac{4}{3}\xi^{3/2} \cdot 246 \text{ GeV} = 0.505 \text{ MeV}$, measured value 0.511 MeV, -1.2% ; cf. the lepton table in Section .7.)

Overall status Doc. 322

The Hilbert-space embedding of FFGFT is structurally complete: $\hat{F}$ is defined, the state space is constructed, all mass formulae from Doc. 320 are identified as eigenvalue equations, and the self-adjointness of $\hat{F}$ is proved on the $\mathbb{Z}_3$ -invariant sector P_0 (Doc. 327, R86, R145): $\xi = \lambda_{\min}(\hat{F}_{D_4})$ is thereby well-defined as a spectral value,

Table 1: All spectral states of the FFGFT and their Hilbert-space components (Doc. 322)

Sector	State	Eigenvalue	Formula	$\mathcal{H}$ -component	Status
Leptons	$ e\rangle$	0.505 MeV	$\frac{4}{3}\xi^{3/2}\nu$	$\mathcal{H}_{\text{fl}}(3, 2)$	[K]
	$ \mu\rangle$	104.96 MeV	$\frac{16}{5}\xi^1\nu$	$\mathcal{H}_{\text{fl}}(5, 4)$	[K]
	$ \tau\rangle$	1783.4 MeV	$\frac{25}{9}\xi^{2/3}\nu$	$\mathcal{H}_{\text{fl}}(9, 5)$	[K]
Neutrinos	$ \nu_1\rangle$	0.976 meV	$m_e\xi^{9/4}$	$\text{Fix}(P) \cap F_2$	[K]
	$ \nu_2\rangle$	9.084 meV	$m_e\xi^2$	$\text{Fix}(P) \cap F_5$	[K]
	$ \nu_3\rangle$	54.11 meV	$m_e\xi^{9/5}$	$\text{Fix}(P) \cap F_7$	[K]
Bosons	$ g_a\rangle$	0	$\hat{F} g_a\rangle = 0$	$\text{Fix}(P_{\mathbb{Z}_3})$	[K]
	$ W^\pm\rangle$	80.38 GeV	$\frac{1}{2}g_2\nu$	$\mathcal{H}_{\text{spin}}$	[B]
	$ Z^0\rangle$	91.19 GeV	$m_W/\cos\theta_W$	$\mathcal{H}_{\text{spin}}$	[B]

provided the D_4 sector lies in P_0 . (Corrected on 1 Oct. 2026: previously “the self-adjointness of $\hat{F}$ is proved” – by Doc. 327, Theorem 3 (corrected), $\hat{F}$ is normal under the $\mathbb{Z}_3$ pairing; it is self-adjoint on P_0 , on the full space only $\text{Re } \hat{F}$ (a choice).) The particle spectrum follows from spectral theory, not from free parameters.